

FLEXHAND · REQUIREMENTS

FlexHand requirements

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DOCUMENT CONTROL

Document ID	FXH-REQ-001	Version	0.3
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/flexhand		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-09-25	Amish Chadha	TRL 3 status from FXH-CAL-001; decisions of FXH-DDR-001 applied (design load kept, R7 target kept pending therapist review, 20 mm cuff limit)
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with concept status
0.1	2026-09-24	Amish Chadha	Initial scaffold

FlexHand requirements

These requirements are checked by calculation in FXH-CAL-001. Five are met on paper, three are at risk, two are not met (R7 pack mass and R11 cuff pressure) and three cannot be verified until TRL 4. The targets are unchanged from v0.2: Amish decided on 2026-09-25 to keep the 30 N design load (FXH-DDR-001, D5) and to keep the 450 g pack target until a therapist has been consulted (D1). Proposed changes to R2, R5 and R11 raised by the calculations are listed in FXH-DDR-001 (N1 to N5) and are not applied here.

Table 1. Requirements and TRL 3 status.

ID	Requirement	Target	Verification	TRL 3 status (FXH-CAL-001)
R1	Move the four fingers through a useful range	MCP 0 to 70°, PIP 0 to 90°, within therapist-set limits	Tendon excursion calculation; later bench test on a finger model	Met: 24.8 mm excursion needed, 29.8 mm stroke
R2	Extend fingers against moderate flexor tone	30 N extensor tendon force per finger at the finger	Torque and friction calculation; published stiffness data	At risk: 0.467 N·m peak at the spool, 119 % of the gearbox's recommended continuous load (RMS 65 %); 30 N covers mild tone (MAS 1+) only, MAS 3 needs about 148 N
R3	Limit force on the hand	Software limit 40 N per finger from motor current; mechanical breakaway at 60 N or less per tendon	Calculation; later bench test	At risk: current senses a finger pair, so one finger can reach 80 N before the pair limit trips; friction spreads the estimate from 32 % below to 18 % above the set value
R4	Deliver a high repetition dose	300 or more full flexion and extension cycles in a 60 min session	Speed calculation	Met: 448 cycles per hour at the fastest setting; holds for strokes up to 5 s
R5	Move slowly and smoothly	Stroke time adjustable from 3 to 15 s (joint speed about 6 to 30 °/s); dwell at each end adjustable	Speed calculation; firmware sketch	At risk: fastest stroke at design load 3.3 s (2.6 s unloaded)
R6	Run several sessions per charge	3 or more 60 min sessions at design load	Power budget	Met: 3.9 sessions at 3.69 Wh each
R7	Light enough to wear seated	Hand-side parts 120 g or less; forearm pack 450 g or less (pack target to be revisited with a therapist, D1)	Mass estimate, then weighing	Not met: hand 103 g (met); pack 675 g
R8	Quick to put on and take off	A care partner dons it in 5 min or less and removes it in 1 min or less	Design review with users	Not verifiable at TRL 3; at risk on flexed, spastic fingers
R9	Stop and release on demand	Physical stop cuts motor power within 100 ms; tendons released by hand in 10 s or less without tools	Calculation; later bench test	Not verifiable at TRL 3: stop in about 11 ms by calculation; release time needs a bench test
R10	Fit most adult hands	Hand length 170 to 205 mm with three glove sizes and adjustable cuffs	Anthropometric check	Met on paper: sizes scaled 0.95, 1.00 and 1.15 from the model
R11	Keep contact pressure tolerable	Mean pressure under a finger cuff 50 kPa or less at design load (proposed limit, to be set with a therapist)	Contact area calculation	Not met: 46 to 84 kPa; anchor cuffs limited to 12.2 to 18.6 mm by the middle phalanx length
R12	Low cost and buildable	Parts cost \$500 or less; no custom PCB for the first build	Priced BOM	Met: \$281.90

ID	Requirement	Target	Verification	TRL 3 status (FXH-CAL-001)
R13	Record what was done	Device logs session time, cycle count and peak motor current; therapist limits cannot be changed from the user controls	Design review	Not verifiable at TRL 3: storage ample (1 MB holds decades of summaries)

■ Assumptions

- Design load: 30 N of extensor tendon force per finger at full extension (D5). Checked against the finger stiffness measured by [Heung et al., *Frontiers in Bioengineering and Biotechnology*, 2020](#): it covers a finger with MAS 1+ tone with a margin of about 1.7, not a finger with MAS 3 tone. It is consistent in scale with the Columbia orthosis, whose single motor provides about 100 N peak for four fingers ([Park et al., arXiv 1802.06131](#)).
- Tendon excursion for 70° at the MCP and 90° at the PIP is 24.8 mm, taking dorsal moment arms of 10 mm and 8 mm; the stroke adds 5 mm for cuff compliance.
- Transmission efficiency is 0.694: capstan friction with a coefficient of 0.1 over 180° of sheath bend, and one idler at 95 %.
- The forearm rests on a support during sessions. The device is not designed to be worn while walking.
- Sessions are supervised by a care partner, at least until a therapist agrees otherwise.

Safety: R3, R9 and R11 are safety requirements. A design that misses any of them must not be worn by a person, even at TRL 4 bench stage. R11 is not met and R3 is at risk.